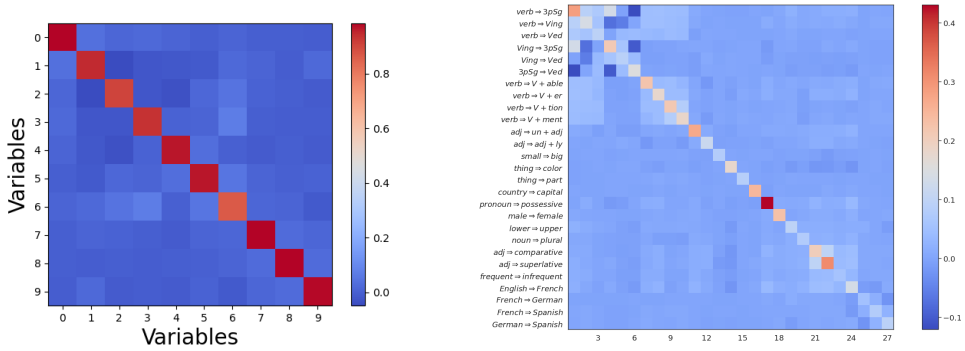

(a) Simulated Experiments

(b) LLaMA-2

Figure 7: The pattern of linear and orthogonal representations matches between simulated experiments and LLaMA-2. Specifically, unembedding steering vectors of the same concept in LLaMA-2 have nontrivial alignment, while steering vectors of different concepts are represented almost orthogonally. The heatmap shows the average cosine similarities between different sets of steering vectors for both simulated experiments and LLaMA-2. For simulated experiments, the cosine similarities are averaged over 10 runs.

F Experiments with large language models

In this section, we will provide additional details on our experiments with LLMs.

Examples of counterfactual context and token pairs Example context pairs from OPUS Books [Tie12] that were used to construct the embedding vectors are shown in Table 5. For the unembedding vectors, we reuse the 27 concepts considered in [PCV23] built atop the work of [GDM16], and they are listed in Table 6 along with example token pairs.

Winograd Schema We now present full details about our experiments on the Winograd Schema dataset [LDM12]. Recall that we consider context pairs arising from the Winograd Schema, which is a dataset of pairs of sentences that differ in only one or two words and which further contain an ambiguity that can only be resolved with world knowledge and reasoning. Example context pairs from the Winograd Schema [LDM12] are shown in Table 7. Note the final 2 context pairs in the table, which have ambiguous concepts, thus highlighting the nuances of this dataset as well as of natural language. We filter the dataset to have the ambiguous word near the end of the sentence to better align with our theory.

We again compute the unembedding vectors for these context pairs output by LLaMA-2. For the embedding vectors, since there is no predefined set of concepts to consider (for instance consider the ambiguous examples), we therefore take the difference of the embedding vectors for the first token that differs in these corresponding pairs of sentences. We compute all pairwise cosine similarities. We observe that for non-matching contexts, the average similarity is 0.011 with a maximum of 0.081, whereas for matching contexts, the average similarity is 0.042 with a maximum of 0.161. This aligns with our predictions that the embedding vectors align better with the unembedding vectors of the same concept.

As an additional experiment, we compute similarities between these Winograd context pairs and the 27 concepts from [PCV23] described above. In Table 8, we display the top 3 pairs of contexts and concepts that have the highest similarities. The alignment seems reasonable as baby, woman, male, and female are different attributes of a person; and wide vs narrow and short vs tall can be construed as different manifestations of small vs big.

Table 5: Example language context pairs from OPUS Books

Language pair	Context 1	Context 2
French–Spanish	Quinze ou seize, répliqua l'autre	Quince ó diez y seis, replicó el otro
French–German	Comment est-il mon maître?	Wie ist er mein Herr?
English–French	I hesitated for a moment.	J'hésitai une seconde.
German–Spanish	Ich hasse die Spazierfahrten	No me gusta salir en coche.

Table 6: Concepts and example token pairs, taken from [PCV23]

Concept	Example token pair	Concept	Example token pair
verb ⇒ 3pSg	(accept, accepts)	verb ⇒ Ving	(add, adding)
verb ⇒ Ved	(accept, accepted)	Ving ⇒ 3pSg	(adding, adds)
Ving ⇒ Ved	(adding, added)	3pSg ⇒ Ved	(adds, added)
verb ⇒ V + able	(accept, acceptable)	verb ⇒ V + er	(begin, beginner)
verb ⇒ V + tion	(compile, compilation)	verb ⇒ V + ment	(agree, agreement)
adj ⇒ un + adj	(able, unable)	adj ⇒ adj + ly	(according, accordingly)
small ⇒ big	(brief, long)	thing ⇒ color	(ant, black)
thing ⇒ part	(bus, seats)	country ⇒ capital	(Austria, Vienna)
pronoun ⇒ possessive	(he, his)	male ⇒ female	(actor, actress)
lower ⇒ upper	(always, Always)	noun ⇒ plural	(album, albums)
adj ⇒ comparative	(bad, worse)	adj ⇒ superlative	(bad, worst)
frequent ⇒ infrequent	(bad, terrible)	English ⇒ French	(April, avril)
French ⇒ German	(ami, Freund)	French ⇒ Spanish	(année, año)
German ⇒ Spanish	(Arbeit, trabajo)		

Table 7: Example context pairs from Winograd Schema

Contexts
The delivery truck zoomed by the school bus because it was going so fast. The delivery truck zoomed by the school bus because it was going so slow.
The man couldn't lift his son because he was so weak. The man couldn't lift his son because he was so heavy.
Joe's uncle can still beat him at tennis, even though he is 30 years younger. Joe's uncle can still beat him at tennis, even though he is 30 years older.
Paul tried to call George on the phone, but he wasn't successful. Paul tried to call George on the phone, but he wasn't available.
The large ball crashed right through the table because it was made of steel. The large ball crashed right through the table because it was made of styrofoam.

Additional barplots The entire set of similarity barplots for the concepts of French–Spanish, French–German, English–French and German–Spanish are in Figures 8, 9, 10 and 11 respectively. As we see, they satisfy the same behavior as described earlier in Section 5.2, exhibiting relatively high similarity with the matching unembedding vector, close to high similarity with related language concepts and low similarity with unrelated concepts.

Table 8: Top similarities between Winograd contexts and token concepts

Contexts	Most similar concept	Similarity
Anne gave birth to a daughter last month. She is a very charming woman. Anne gave birth to a daughter last month. She is a very charming baby .	male⇒female	0.311
The table won't fit through the doorway because it is too wide. The table won't fit through the doorway because it is too narrow.	small⇒big	0.309
John couldn't see the stage with Billy in front of him because he is so short. John couldn't see the stage with Billy in front of him because he is so tall.	small⇒big	0.303

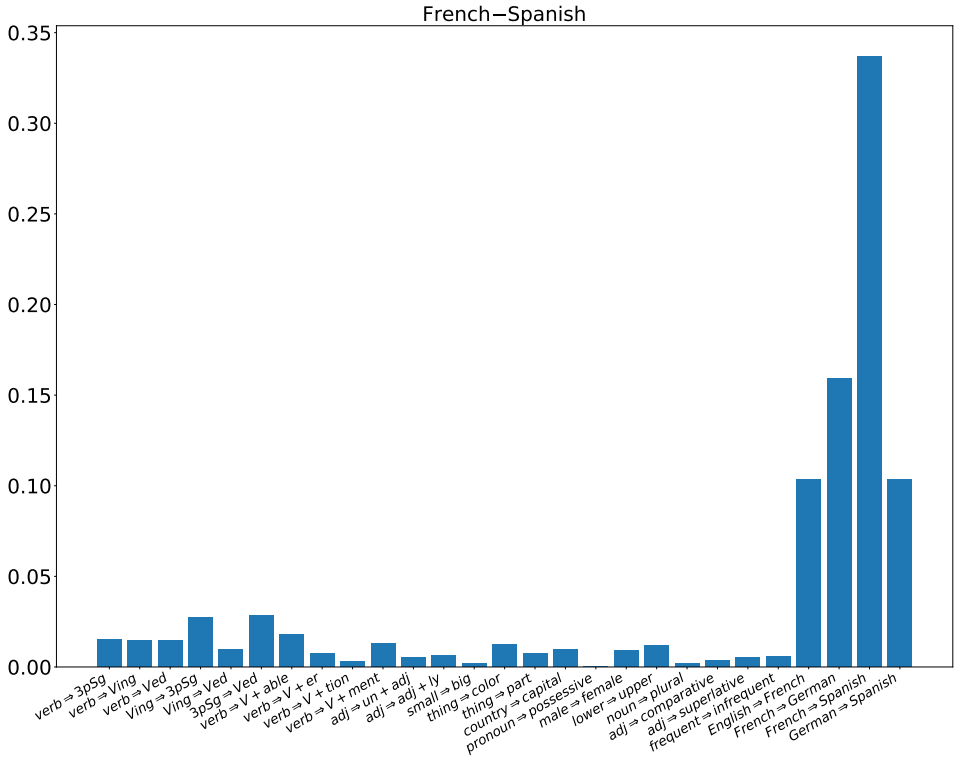

Figure 8: The French–Spanish concept is highly correlated with similar token concepts relative to others. This figure shows all cosine similarities between the French–Spanish concept and token concepts.